Aptitude Formula Revision Sheet

1. Trains

Speed = Distance / Time

Convert km/hr to m/s = km/hr × (5/18)

Convert m/s to km/hr = m/s × (18/5)

Time to cross pole = Length of train / Speed

Time to cross platform = (Length of train + platform) / Speed

Time to cross another train (opposite) = (Sum of lengths) / (Sum of speeds)

Time to cross another train (same) = (Sum of lengths) / (Difference of speeds)

2. Boats and Streams

Downstream speed = Boat speed + Stream speed

Upstream speed = Boat speed - Stream speed

Time = Distance / Speed

If it takes T1 hours downstream and T2 hours upstream for same distance D:

Boat speed = D × (T2 + T1) / (2 × T1 × T2)

Stream speed = D × (T2 - T1) / (2 × T1 × T2)

3. Problems on Ages

Present age = x

Age after n years = x + n

Age n years ago = x - n

Use given ratios or differences to form equations

4. Time, Speed, and Distance

Speed = Distance / Time

Distance = Speed × Time

Time = Distance / Speed

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Relative speed (same direction) = Faster - Slower

Relative speed (opposite direction) = Sum of speeds

5. Pipes and Cisterns

If pipe A fills in x hours, B fills in y hours, together = $1/x + 1/y$

If one empties: Net = $1/x - 1/y$

Time to fill = 1 / Net rate

6. Time and Work

Work = Efficiency × Time

Efficiency = 1 / Time taken

If A does x in a days, B in b days, together = $1/x + 1/y$

Work = Men × Days × Hours (constant)

7. Profit, Loss, and Discount

Profit = SP - CP, Loss = CP - SP

Profit% = (Profit / CP) × 100

Loss% = (Loss / CP) × 100

SP = CP × (1 ± Profit%/100)

CP = SP / (1 ± Profit%/100)

Marked Price = SP / (1 - Discount%/100)

8. Simple and Compound Interest

Simple Interest = $(P \times R \times T) / 100$

Compound Interest = $P(1 + R/100)^T - P$

Amount = $P(1 + R/100)^T$

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9. Area, Surface Area and Volume

Rectangle: l × b

Square: a^2

Circle: pir^2, Circumference: 2pir

Triangle: ½ × b × h

Sphere Volume: (4/3)pir^3, Surface: 4pir^2

Cylinder Volume: pir^2h, Surface: 2pir(h + r)

Cone Volume: (1/3)pir^2h, Surface: pir(l + r)

Cube Volume: a^3, Surface: 6a^2

Cuboid Volume: l × b × h, Surface: 2(lb + bh + hl)

10. Ratio and Proportion

Ratio: a:b = a/b

Proportion: a:b = c:d => a/b = c/d

If a:b and b:c are given, a:b:c = Multiply common term

11. Averages

Average = Sum of observations / Number of observations

Sum = Average × Number of observations

12. Mixtures and Alligations

Alligation Rule: (High - Mean):(Mean - Low) = Quantity of low:high

Mean = (qty1 × rate1 + qty2 × rate2) / total quantity

13. Permutations and Combinations

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nPr = n! / (n - r)!

nCr = n! / [r!(n - r)!]

Arrangements with repetition = n^r

14. Probability

Probability = Favourable outcomes / Total outcomes

P(A or B) = P(A) + P(B) - P(A and B)

P(not A) = 1 - P(A)